Depth First Search (DFS):

The aim of DFS algorithm is to traverse the graph in such a way that it tries to go far from the root node. Stack is used in the implementation of the depth first search.

Algorithm:

1. Push the root node in the Stack.
2. Loop until stack is empty.
3. Peek the node of the stack.
4. If the node has unvisited child nodes, get the unvisited child node, mark it as traversed and push it on stack.
5. If the node does not have any unvisited child nodes, pop the node from the stack.

Program:

```
void DFS(int);
int G[10][10],visited[10],n;
void main(){
int i,j; printf("Enter number of vertices:");
scanf("%d",&n);
printf("\n Enter adjency matrix of the graph:");
for(i=0; i<n; i++)
for(j=0; j<n; j++)
scanf("%d", &G[i][j]);
for(i=0; i<n; i++)
visited[i] = 0;
DFS(0); }
```

```
void DFS (int i)
{ int j;
printf("\n%d",i);
visited [i] = 1;
for(j=0; j<n; j++)
 if(!visited [j] && G[i][j]==1)
  DFS (j);
}
```

<u>Output:</u>

Enter number of vertex : 4

Enter adjency matrix of the graph

```
1   0   1   1
0   0   1   0
0   1   0   1
1   0   1   0
```